

3D Character Motion Authoring From Temporally-Sparse Photos

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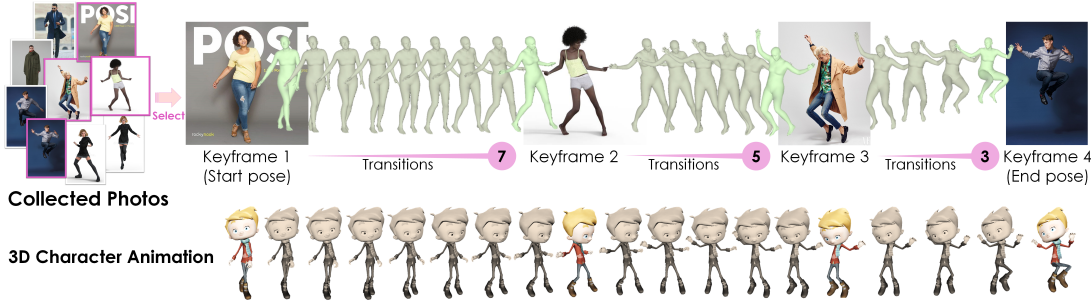


Figure 1: Our system generates pose and shape transitions in-between given photos (keyframes) to create 3D character motion. Bright colors represent the generated pose and shape of the keyframes, while dark colors represent the generated transitions.

ABSTRACT

This paper presents a neural network-based learning approach that enables seamless generation of 3D human motion in-between photos to accelerate the process of 3D character motion authoring. This new approach allows users to freely edit (replace, insert, or delete) input photos and specify the transition length to generate a kinematically coherent sequence of 3D human poses and shapes in-between the given photos. We demonstrate through qualitative and subjective evaluations that our approach is capable of generating high-fidelity, natural 3D pose and shape transitions.

ACM Reference Format:

Jen-Chun Lin and Wen-Li Wei. 2023. 3D Character Motion Authoring From Temporally-Sparse Photos. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '23 Posters)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588028.3603656>

1 INTRODUCTION

Convincing performances for computer-generated (CG) characters are vital in the game and animation industries. The keyframing process, a primary technique for creating animations, remains labor-intensive and challenging despite advancements in tools and methods. Animators still need to adhere to detailed pose specifications to ensure naturalness in drawing body motion in-between keyframes. Although advanced motion capture (MOCAP) systems can outperform keyframing techniques in creating CG characters with numerous degrees of freedom, they still necessitate specialized

equipment and manual post-processing, leading to expensive animations. As a result, developing a data-driven approach that leverages existing 3D and 2D data to automatically generate kinematically coherent 3D human motion in-between keyframes is advantageous for producing CG character animations.

In this work, we aim to generate 3D poses and shapes in-between photos (see Figure 1), based on a data-driven approach. This enables animators or novice users to create CG character animations from the resulting 3D pose and shape transitions by simply taking or searching for photos that correspond to the actions they envision. Pertinent to our task are 3D human pose and shape estimation [Kolotouros et al. 2019], as well as in-betweening [Harvey et al. 2020]. The former aims to estimate the 3D pose and shape of individuals from each input RGB image, while the latter generates in-between 3D poses by relying on 3D joint-based keyframes and context information from preceding frames as animation constraints. For example, in the case of the in-betweening task, the system requires novice users to provide 3D joint-based keyframes that include context from preceding frames, which is impractical. Unlike the tasks of 3D human pose and shape estimation and in-betweening, our approach takes only temporally-sparse photos (RGB-based keyframes) as animation constraints and generates 3D poses and shapes in-betweening. This can be seen as accomplishing both of the above tasks simultaneously under more stringent conditions. To achieve this, we present a neural network-based learning approach to learn diverse motion transitions from masked action sequences. We show that our approach can generate high-fidelity, customizable 3D pose and shape transitions with adjustable lengths, which greatly streamlines and accelerates the 3D character motion authoring process.

2 OUR APPROACH

Our approach consists of four components, a static feature extractor, a feature interpolator, a temporal feature encoder, and an SMPL parameter regressor (see Figure 2). Given a sequence of temporally-sparse photos $P = \{I_n\}_{n=1}^N$ with N frames, we first use ResNet-50,

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SIGGRAPH '23 Posters, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0152-8/23/08.

<https://doi.org/10.1145/3588028.3603656>

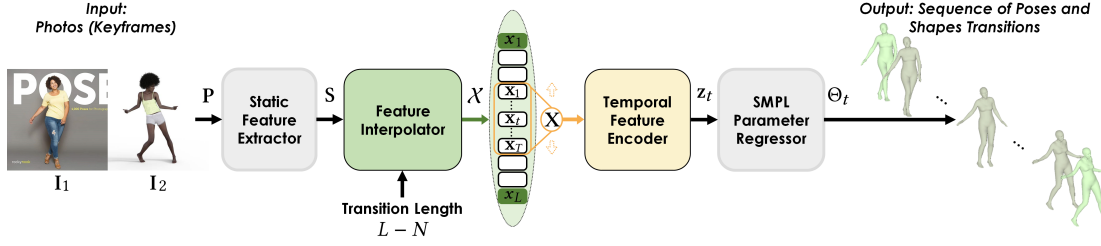


Figure 2: An overview of our approach. We estimate Θ in-between given photos to generate 3D pose and shape transitions.

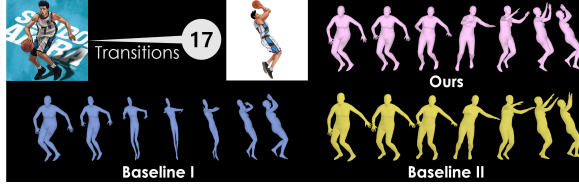


Figure 3: Qualitative comparison: We sample the generated transitions at equal intervals to display the result.

which was pre-trained on a single image-based 3D human pose and shape estimation task [Kolotouros et al. 2019], to extract the static features of each frame to form a static feature representation sequence $S = \{s_n\}_{n=1}^N$. The extracted S is then sent to the feature interpolator to produce an interpolated static feature representation sequence $X = \{x_l\}_{l=1}^L$ with L consecutive frames, where the linear interpolation is used to fill in missing static features in-between temporally-sparse photos along the temporal axis. Note that the user can specify the number of interpolated static features, *i.e.*, transition length $(L - N)$. Next, we sequentially send the interpolated static feature representation sequence $X = \{x_l\}_{l=1}^L$ of sequence length T to the temporal feature encoder to generate temporal feature representation sequence. In the temporal feature encoder, we first use the self-attention mechanism to model the non-local contextual relations, which capture long-range dependencies throughout the sequence with a length of T , thereby refining the interpolated static features into temporal features that better describe human motion. The temporal feature representation sequence is then aggregated using a fully connected feed-forward neural network to obtain an aggregated temporal feature representation z_t for the target missing frame t . Finally, we estimate the pose, shape, and camera parameters $\Theta_t \in \mathbb{R}^{85}$ needed for the parametric SMPL model [Loper et al. 2015] to decode the 3D human mesh based on each aggregated z_t , using the SMPL parameter regressor proposed in [Kolotouros et al. 2019].

Similar to [Kolotouros et al. 2019], we adopt batches of mixed 3D and 2D datasets for training. To simulate various transitions in-between temporally-sparse photos and generate high-quality 3D poses and shapes at specified lengths, we randomly mask 75% of images in each action sequence. Then, we interpolate using the corresponding unmasked images in the latent (static feature) space to fill in the masked regions (images) for network training. With ground-truth annotations for masked images, we impose an $\mathcal{L}_2$ loss between estimated and ground-truth SMPL parameters and 3D/2D

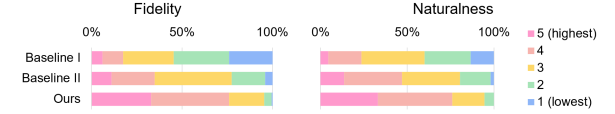


Figure 4: Subjective evaluation of fidelity and naturalness, conducted with 40 participants.

joint coordinates as well as an adversarial loss to guide network learning.

3 EVALUATION AND DISCUSSION

We qualitatively compare our approach to two baselines. Baseline I estimates human meshes from the given photos and directly interpolates them linearly. Baseline II is created by omitting the temporal feature encoder and directly sending the interpolated static features to the SMPL parameter regressor (see Figure 2). The results in Figure 3 show that direct linear interpolation in-between human meshes (Baseline I) often results in unnatural body deformations (paper-like effect), failing to generate high-fidelity 3D pose and shape transitions. While Baseline II leverages the SMPL model for better human mesh decoding, it still faces challenges in accurately producing 3D pose and shape transitions from given photos without employing the temporal feature encoder. For example, the pose is not similar enough to the provided photos. With the support of the temporal feature encoder, our approach is shown to achieve high-fidelity and natural 3D pose and shape transitions. Subjective evaluation with a 5-point opinion score on the 3D pose and shape transitions from three approaches across five sets also showcases the capability of our approach to generate high-fidelity and natural transitions (see Figure 4). Despite encouraging results, we find our current approach falls short in generating transitions for actions beyond the training set; we intend to develop data augmentation techniques to address this issue in future research endeavors.

ACKNOWLEDGMENTS

This work was supported in part by MOST under grant 110-2221-E-001-016-MY3 and Academia Sinica under grant AS-TP-111-M02.

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